ROSWELL PARK
COMPREHENSIVE CANCER CENTER

PATHOLOGY REPORT

Left Axillary Lymph Node

PATIENT: Jane Doe	DATE OF BIRTH: 01/01/1936
MEDICAL RECORD NUMBER: MRN 8745618	GENDER: F
SPECIMEN COLLECTION DATE/TIME: 01/01/201912:00h	CASE NUMBER: aalsk48416
SPECIMEN RECEIPT DATE/TIME: 12/17/2025	

SPECIMEN LABEL(s): A. Lymph node, left axilla, core needle biopsy.

PATIENT HISTORY: history of NSCLC s/p SBRT in 2020 then recurrence

FINAL DIAGNOSIS:

A. LEFT AXILLARY LYMPH NODE, CORE BIOPSY:
- METASTATIC ADENOCARCINOMA, LUNG PRIMARY.
- Morphology and immunoprofile (TTF-1+, Napsin A+, GATA3-) support origin from the patient's known non-small cell lung cancer.
- See separate addendum for molecular profiling.

Gross Description

Part A consists of three linear cores of tan-white fibroadipose tissue measuring 0.8 cm to 1.5 cm in length by 0.1 cm in diameter. Entirely submitted in cassette A1.

Microscopic description:

Sections show cores of lymphoid tissue extensively replaced by a poorly differentiated carcinoma. The tumor cells form irregular glandular structures and solid nests set within a desmoplastic stroma. Cytologically, the cells exhibit pleomorphism, high nuclear-to-cytoplasmic ratio, and prominent nucleoli. Mitotic figures are frequent

Immunohistochemistry:

Immunostains were performed to determine tumor origin (lung vs. breast vs. other).
- TTF-1: Positive (diffuse, strong nuclear staining)
- Napsin A: Positive
- CK7: Positive
- CK20: Negative
- GATA-3: Negative
- Mammaglobin: Negative
- p40: Negative

Breast Prognostic Markers (Performed due to axillary location):
- ER (Estrogen Receptor): Negative
- PR (Progesterone Receptor): Negative
- HER2/neu (IHC): 1+ (Negative/Low expression)

Date and electronically signed: 12/26/2025 by Dr. River Tree

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ADDENDUM: COMPREHENSIVE GENOMIC PROFILING (NGS)

Report Date: January 10, 2026

Test Methodology: Targeted Next-Generation Sequencing (500 gene panel) performed on FFPE tissue.

QUALITY METRICS Tumor Cellularity: 40% Mean Coverage: 650x

BIOMARKER SUMMARY

- Tumor Mutational Burden (TMB): 4 Muts/Mb (Low)
- Microsatellite Status (MSI): Stable (MSS)
- PD-L1 (22C3): TPS <1%

GENOMIC ALTERATIONS DETECTED

Tier I/II Variants (Pathogenic / Likely Pathogenic)

1. TP53 c.524G>A (R175H)
    - Allele Frequency: 22%
    - Interpretation: Loss of function mutation common in NSCLC.
2. ERBB2 c.2326_2327insTGT (A775_G776insYVMA)
    - Allele Frequency: 18%
    - Interpretation: In-frame insertion in Exon 20.

Variants of Uncertain Significance (VUS)

- ATM c.8320G>A (V2774I)
- KMT2D c.401C>T (P134L)
- LRP1B c.1220A>G (N407S)

Pertinent Negatives (Wild Type/Not Detected):

- EGFR (including Exon 19 del, L858R, T790M)
- KRAS (including G12C)
- ALK Fusions
- ROS1 Fusions
- BRAF V600E
- MET Exon 14 skipping
- RET Fusions
- NTRK 1/2/3 Fusions

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